

MODULE 16

SQL Window Functions

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

SQL Window Functions

What are Window Functions?

Calculations over related rows without GROUP BY collapse.

Syntax:

function() OVER (PARTITION BY col ORDER BY col)

ROW_NUMBER RANK DENSE_RANK

```
SELECT first_name, gpa,  
ROW_NUMBER() OVER(PARTITION BY dept ORDER BY gpa DESC) rn,  
RANK() OVER(PARTITION BY dept ORDER BY gpa DESC) rnk,  
DENSE_RANK() OVER(PARTITION BY dept ORDER BY gpa DESC) dr  
FROM students;
```

ROW_NUMBER: 1,2,3,4 unique
RANK: 1,2,2,4 skips after tie
DENSE_RANK: 1,2,2,3 no skip

LAG and LEAD

```
SELECT month, revenue,  
LAG(revenue) OVER(ORDER BY month) AS prev_month,  
LEAD(revenue) OVER(ORDER BY month) AS next_month,  
revenue - LAG(revenue) OVER(ORDER BY month) AS growth  
FROM monthly_sales;
```

LAG = look backward LEAD = look forward

NTILE and PERCENT_RANK

```
SELECT first_name, gpa,  
NTILE(4) OVER(ORDER BY gpa DESC) AS quartile,  
ROUND(PERCENT_RANK() OVER(ORDER BY gpa), 2) AS pct  
FROM students;
```

Running Total

```
SELECT month, revenue,  
SUM(revenue) OVER(  
ORDER BY month ROWS UNBOUNDED PRECEDING  
) AS running_total,  
AVG(revenue) OVER(  
ORDER BY month  
ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
```

```
) AS moving_avg_3  
FROM monthly_sales;
```